

Fraud Detection – Candidate Selection Project

Ajay – Project Notes

1. Data Cleaning

- Dataset size: 6,362,620 rows.
- No missing values.
- No duplicate rows.
- Fraud rate: ~0.13% (highly imbalanced).
- Merchant accounts do not have destination balance info.
- Created features: isMerchant, error_balance_dest_clean.
- Dropped: nameOrig, nameDest, newbalanceOrig, newbalanceDest.

2. Model Used

Tested Logistic Regression and Random Forest.

Logistic Regression: High recall (0.99) but very low precision (0.03).

Random Forest: Precision 0.98, Recall 0.75, ROC-AUC 0.98.

Final Model Selected: Random Forest.

3. Variable Selection Logic

Selected based on business logic and feature importance.

Top variables:

- amount
- error_balance_dest_clean
- oldbalanceOrg
- step
- error_balance_orig
- oldbalanceDest

- type_TRANSFER

- type_CASH_OUT

4. Model Performance

Confusion Matrix (Test Set):

- True Fraud Detected: 97

- Fraud Missed: 32

- False Alarms: 2

Metrics:

- Precision (Fraud): 0.98

- Recall (Fraud): 0.75

- ROC-AUC: 0.98

5. Key Factors Predicting Fraud

- High transaction amount

- Abnormal destination balance behavior

- High origin balance

- Transfer and Cash-Out transactions

- Time pattern (step)

6. Do These Factors Make Sense?

Yes. Fraudsters transfer large amounts and drain accounts.

Model findings align with actual fraud behavior.

7. Recommended Prevention Strategy

- Real-time monitoring of large transfers

- Flag abnormal balance mismatches

- Additional authentication for high-value transfers
- Monitor rapid account draining
- Freeze suspicious accounts automatically

8. How to Measure Effectiveness

- Compare fraud rate before vs after
- Measure financial loss reduction
- Monitor false positive rate
- Track investigation cost
- Monitor model drift over time

Final Conclusion:

Random Forest model provides strong fraud detection with high precision and good recall.

Operationally feasible and aligned with business logic.